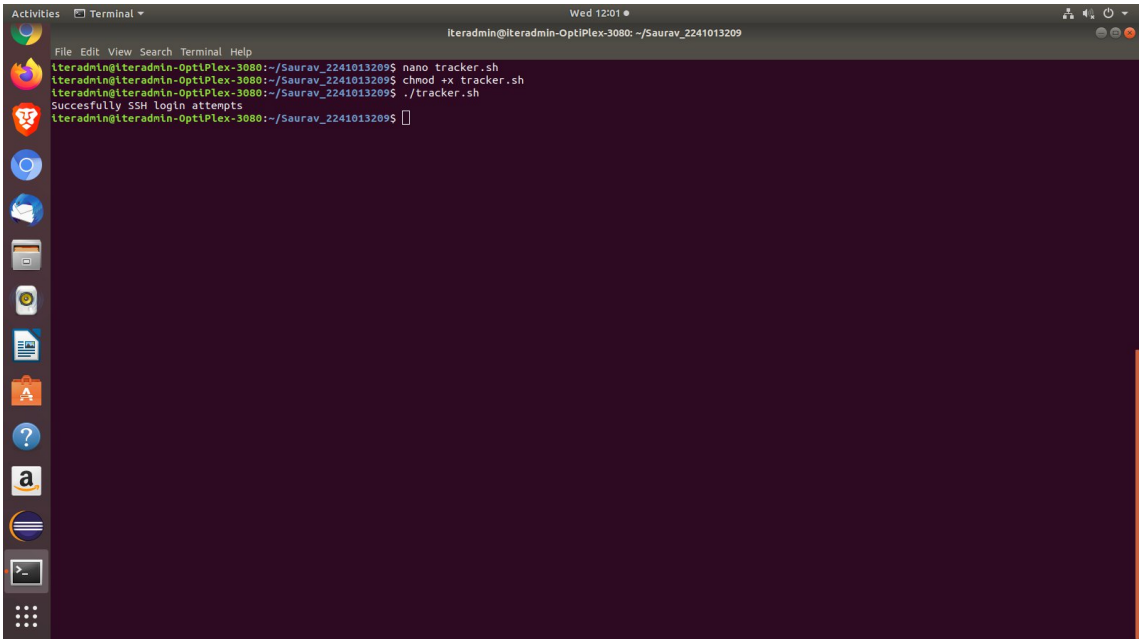
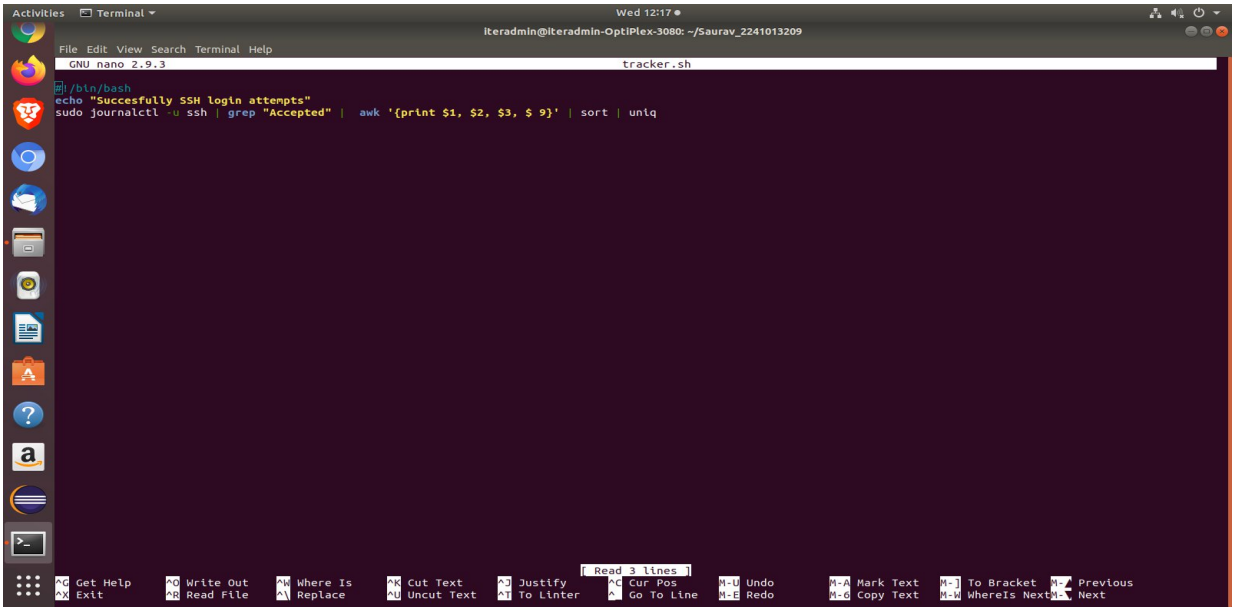
ASSIGNMENT – 1

1. User Login Tracker

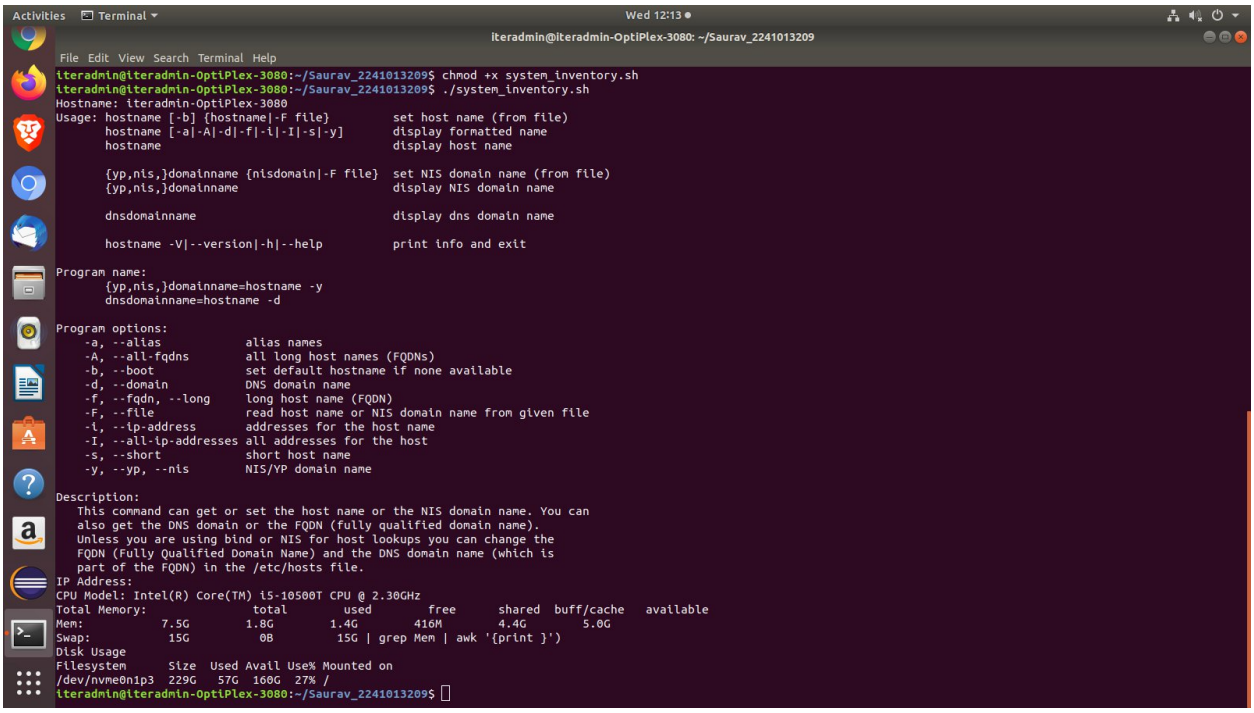
   Write a script to read the /var/log/auth.log file and extract the list of successful SSH login attempts using grep, awk, or cut. Display the unique usernames and their login timestamps.

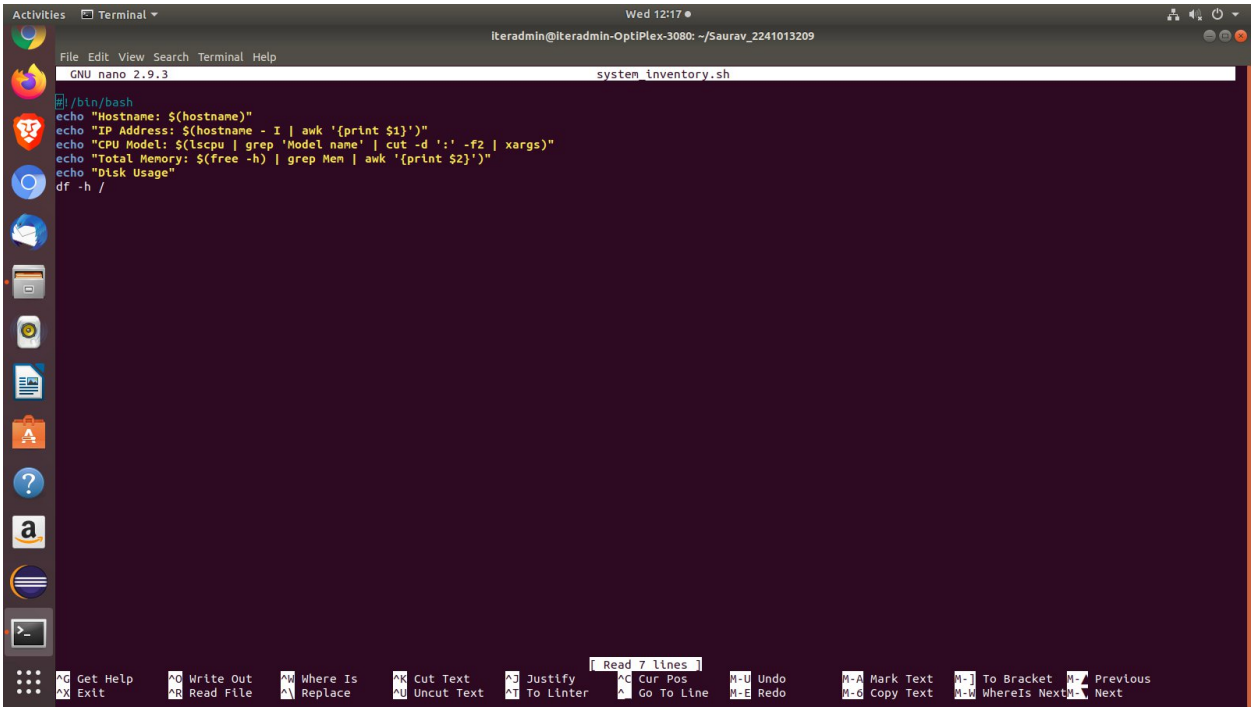




2. System Inventory Report

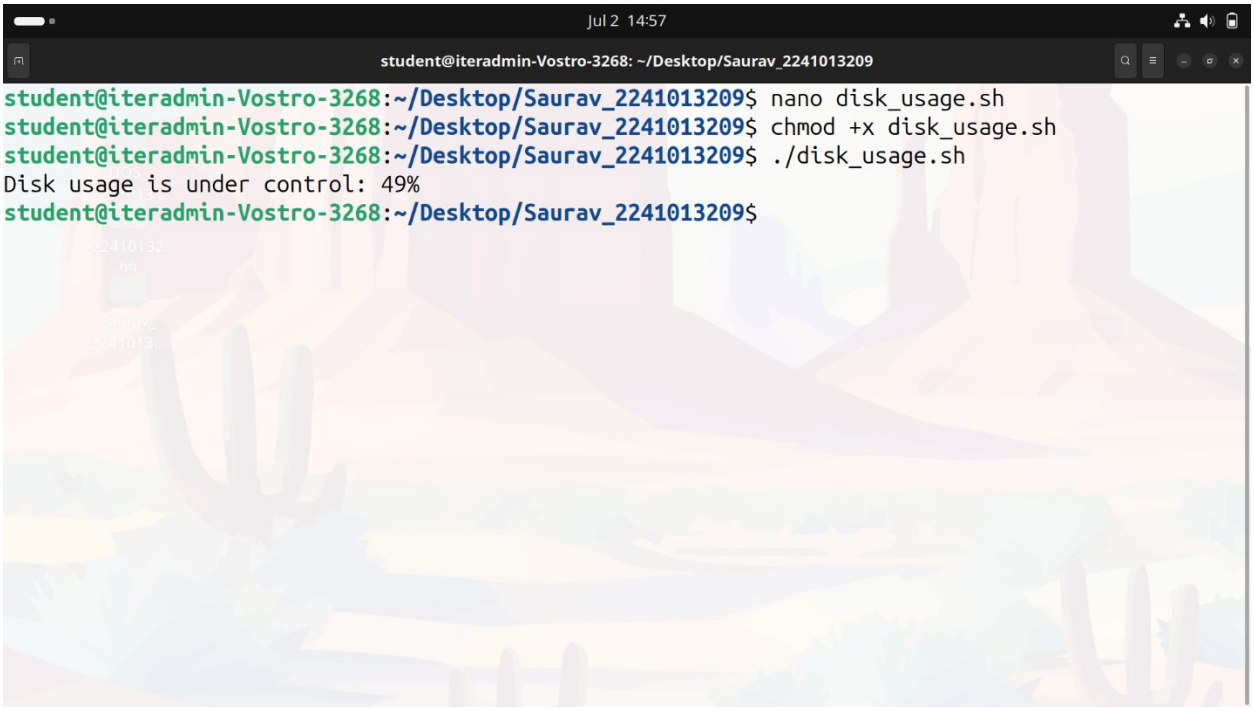
Create a script that gathers and displays system information: hostname, IP address, CPU model, total memory, and disk usage. Use commands like uname, hostname, ifconfig, free, and df.

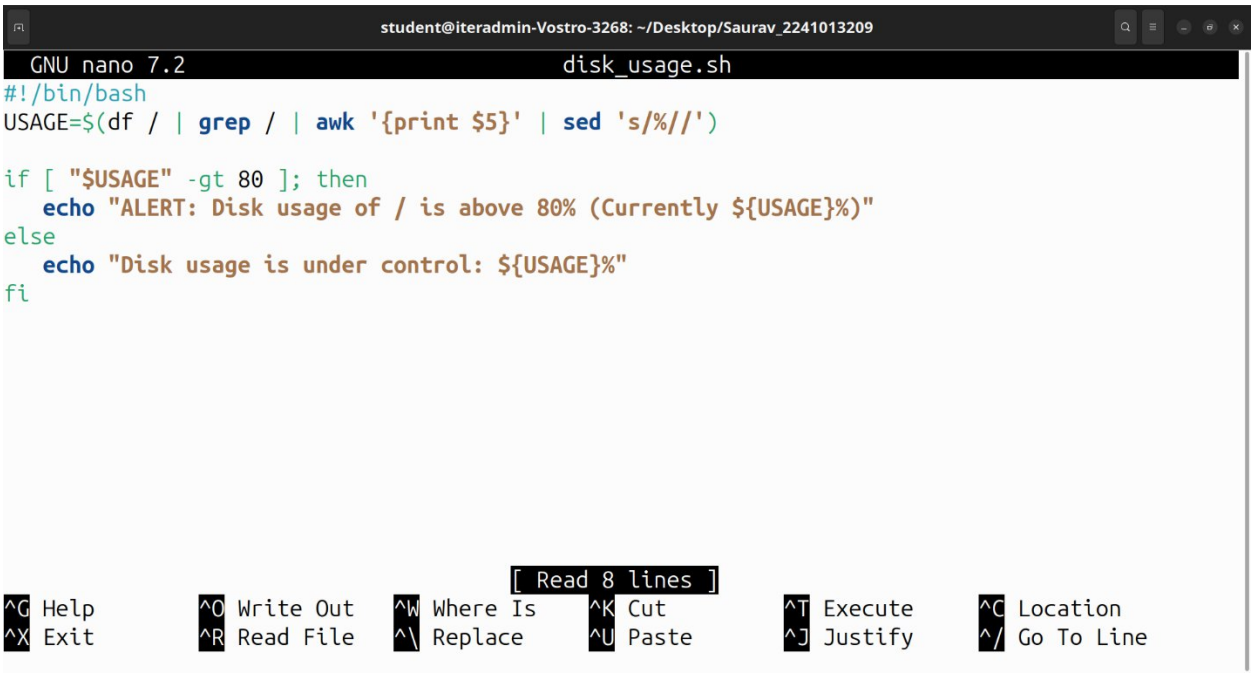




3. Check Disk Usage & Alert

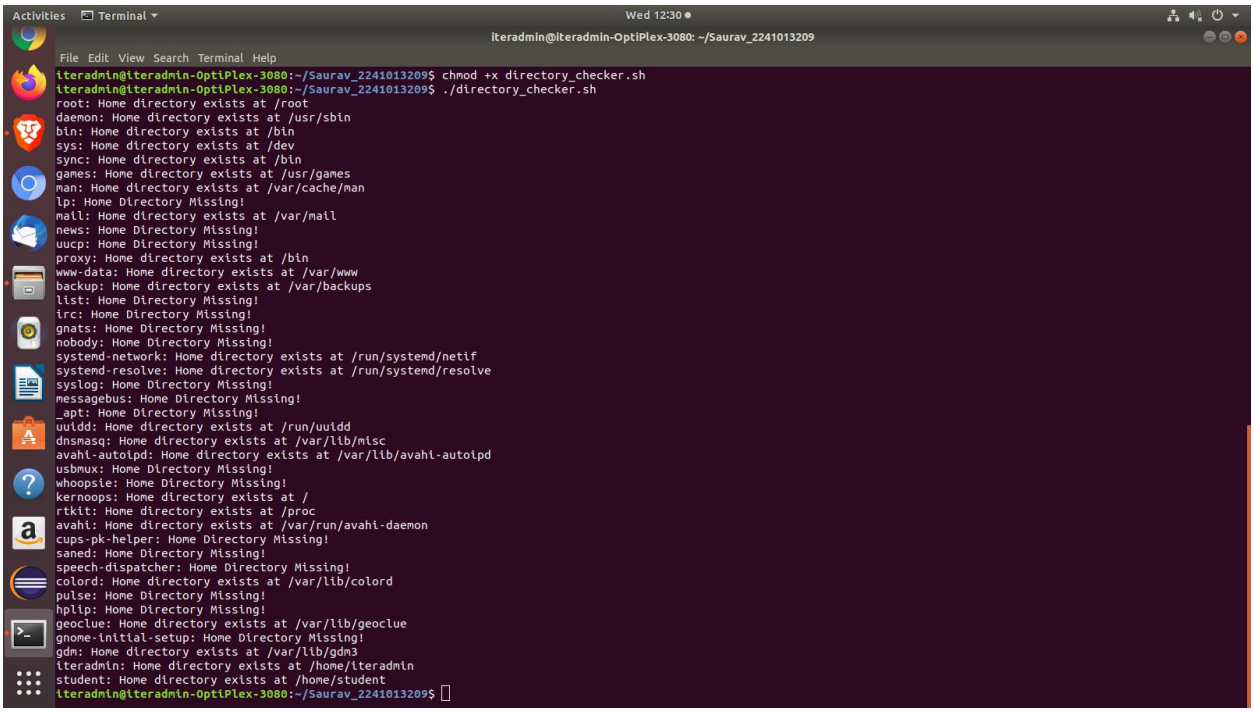
Write a script that checks the disk usage of root (/) partition. If the usage exceeds 80%, print an alert. Use df, awk, and if conditionals.

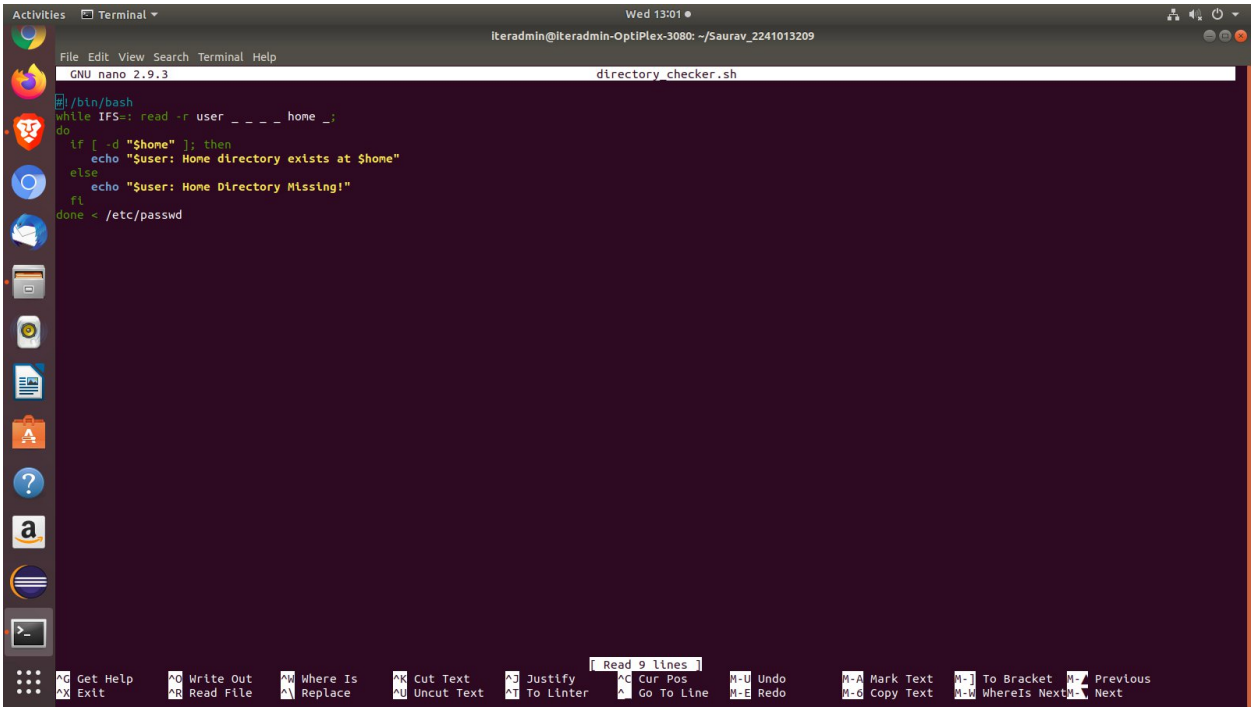




4. User Home Directory Checker

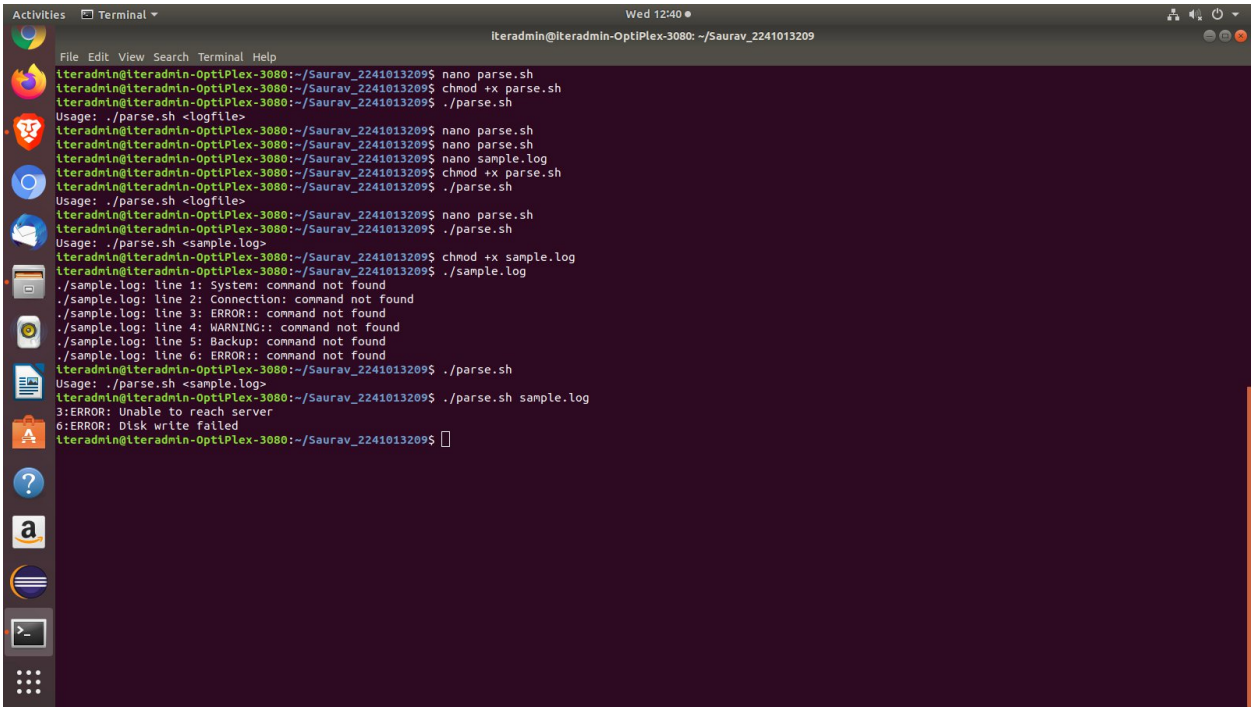
Loop through all users in /etc/passwd, extract their usernames and home directories using cut, and print whether the directory exists or not using an if check.

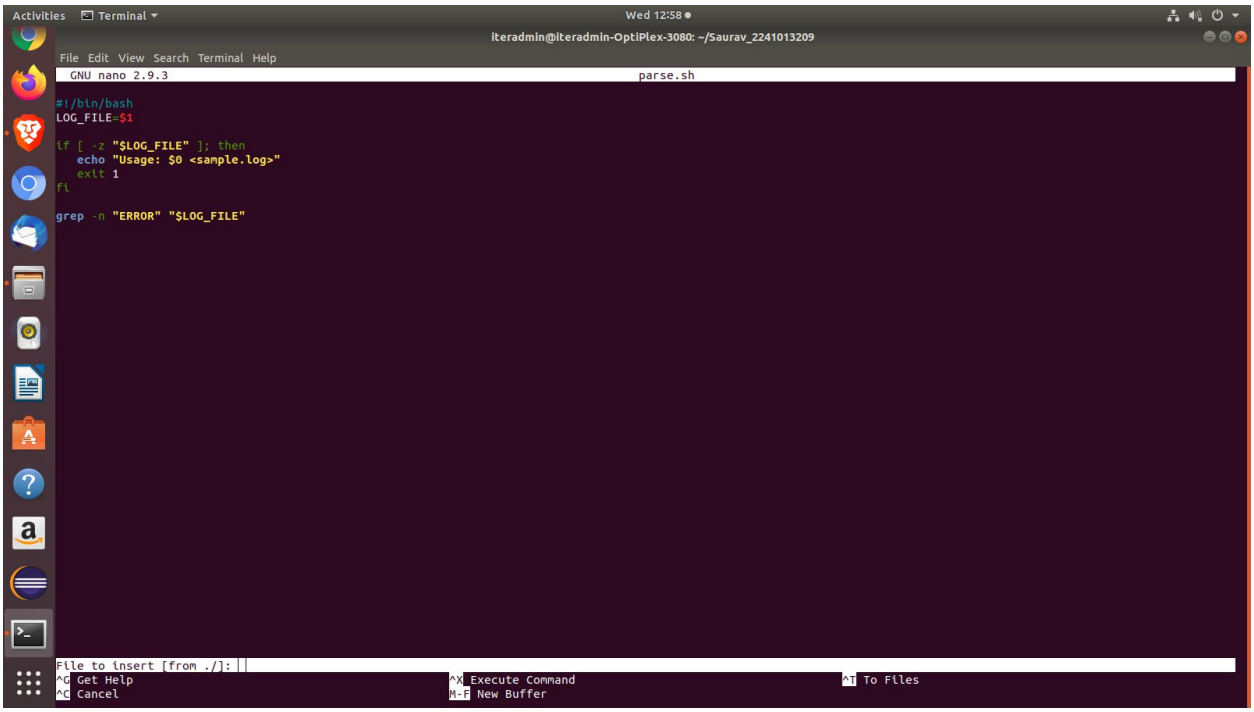




5. Parse and Filter Logs

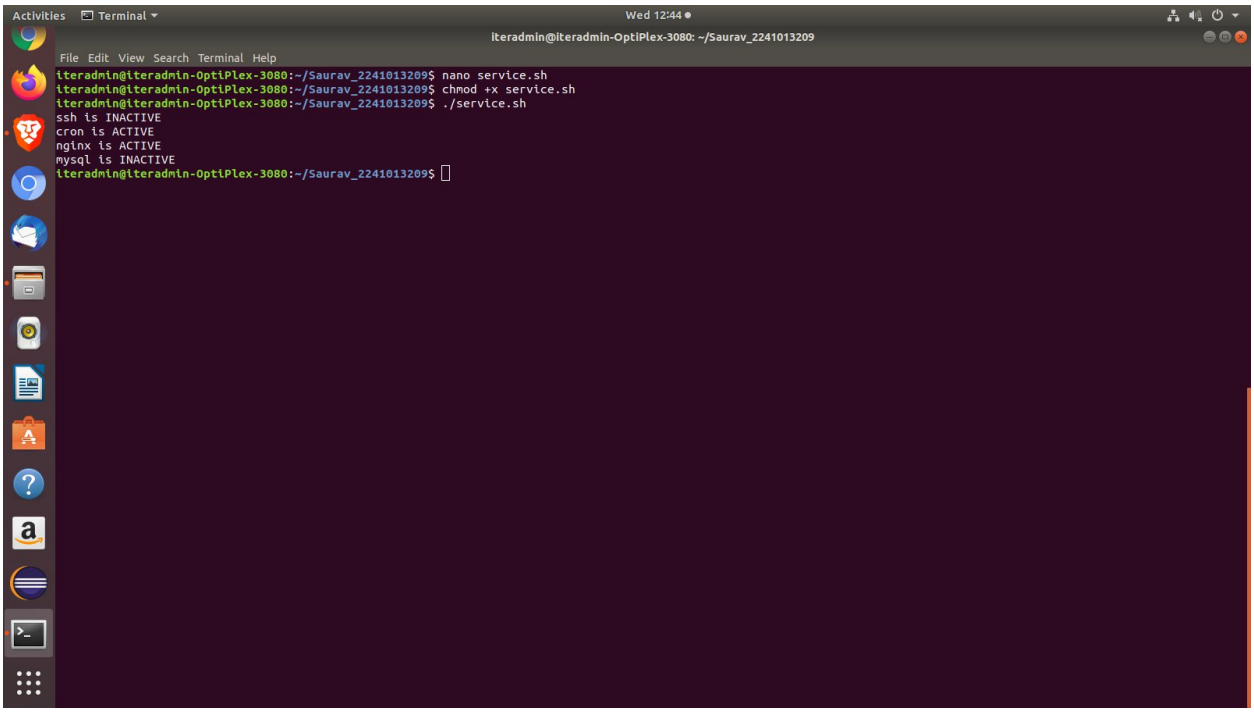
Using grep and awk, write a script to parse a given log file and list all lines containing the word “ERROR” along with the line number.

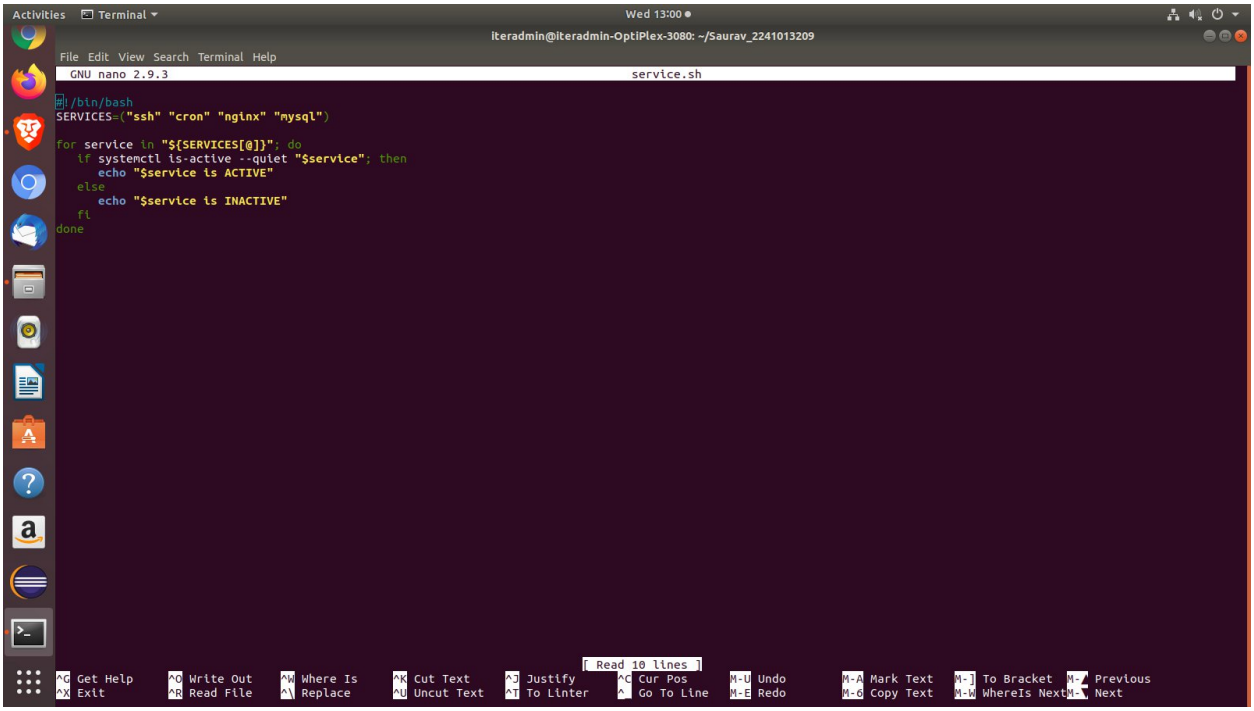




6. Service Status Checker

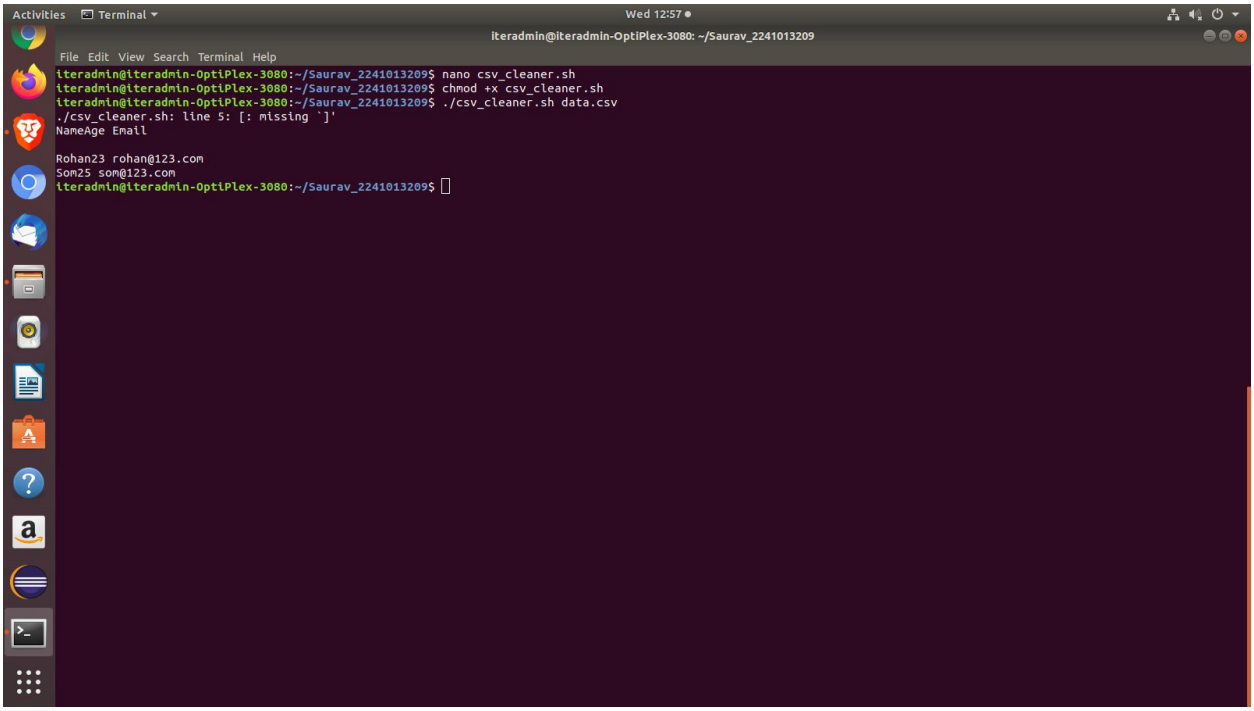
Create a script that takes a list of service names and checks whether each service is active or inactive using systemctl or service. Use loops and if statements.

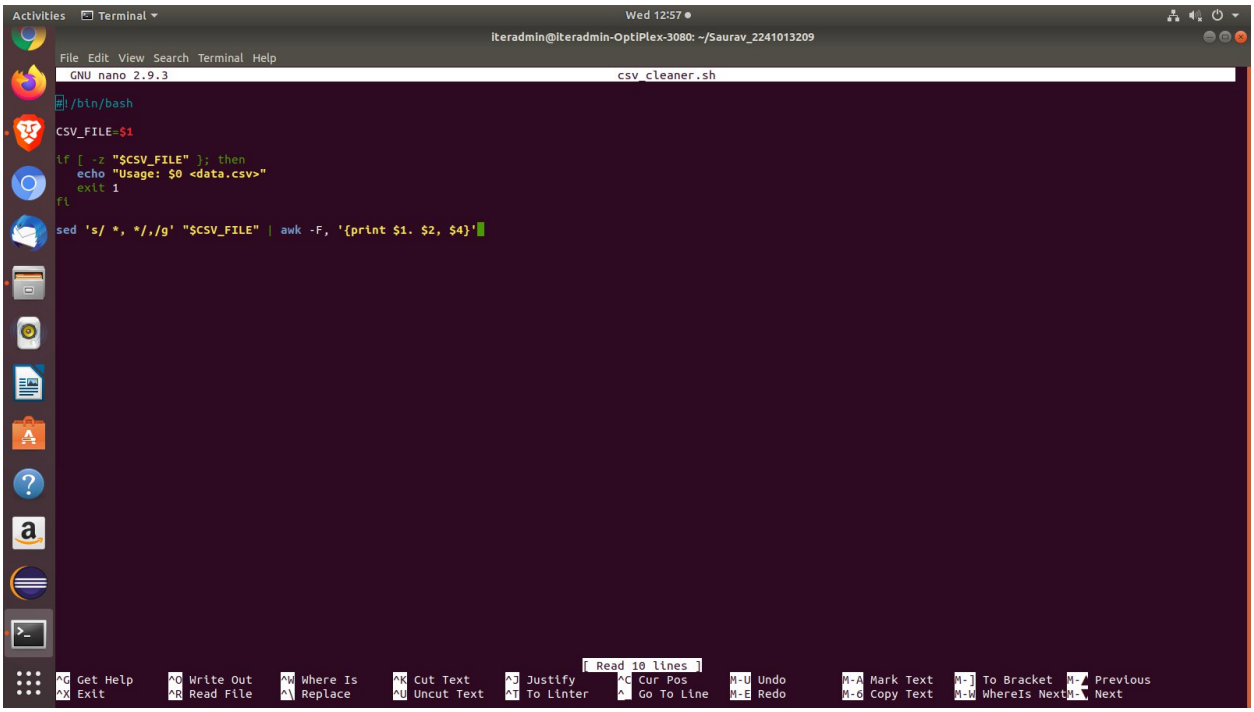




7. CSV Data Cleaner

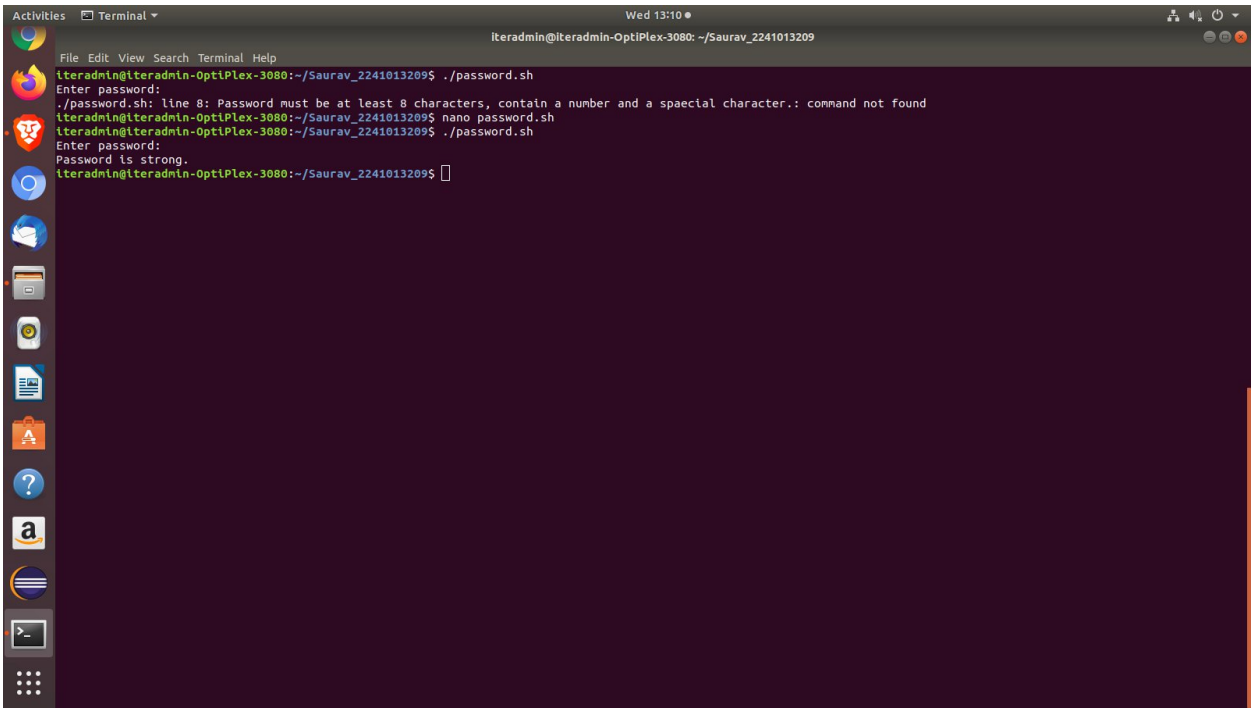
Given a CSV file with extra spaces and inconsistent delimiters, use awk, sed, and cut to clean it and print only selected columns neatly.

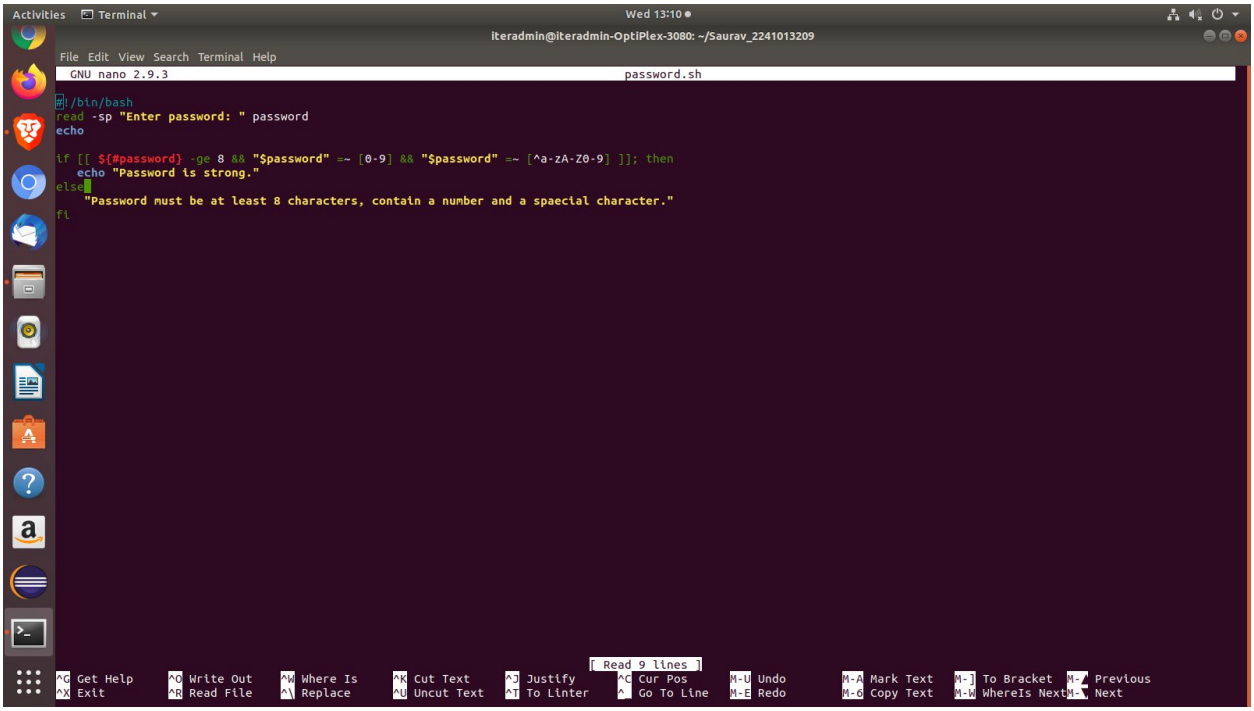




8. Password Strength Validator

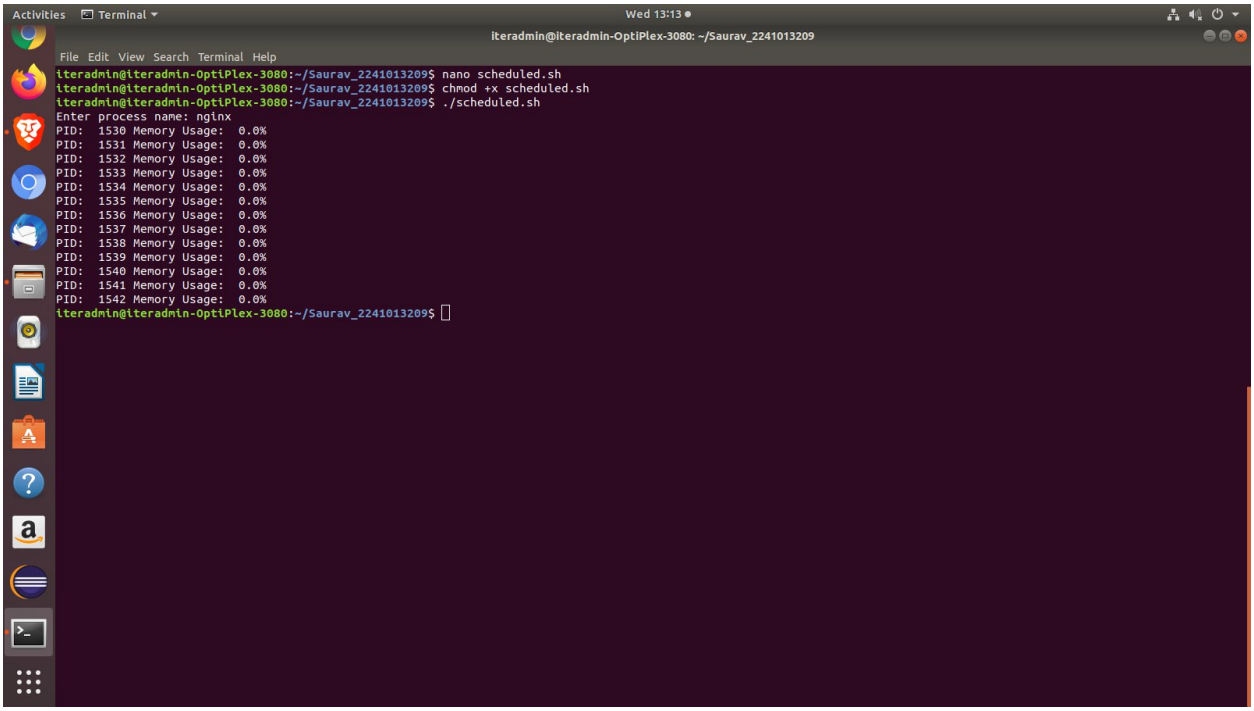
Accept a string as input and use a combination of regex (grep or [[… =~]]) to validate that it’s at least 8 characters, contains one number, and one special character.

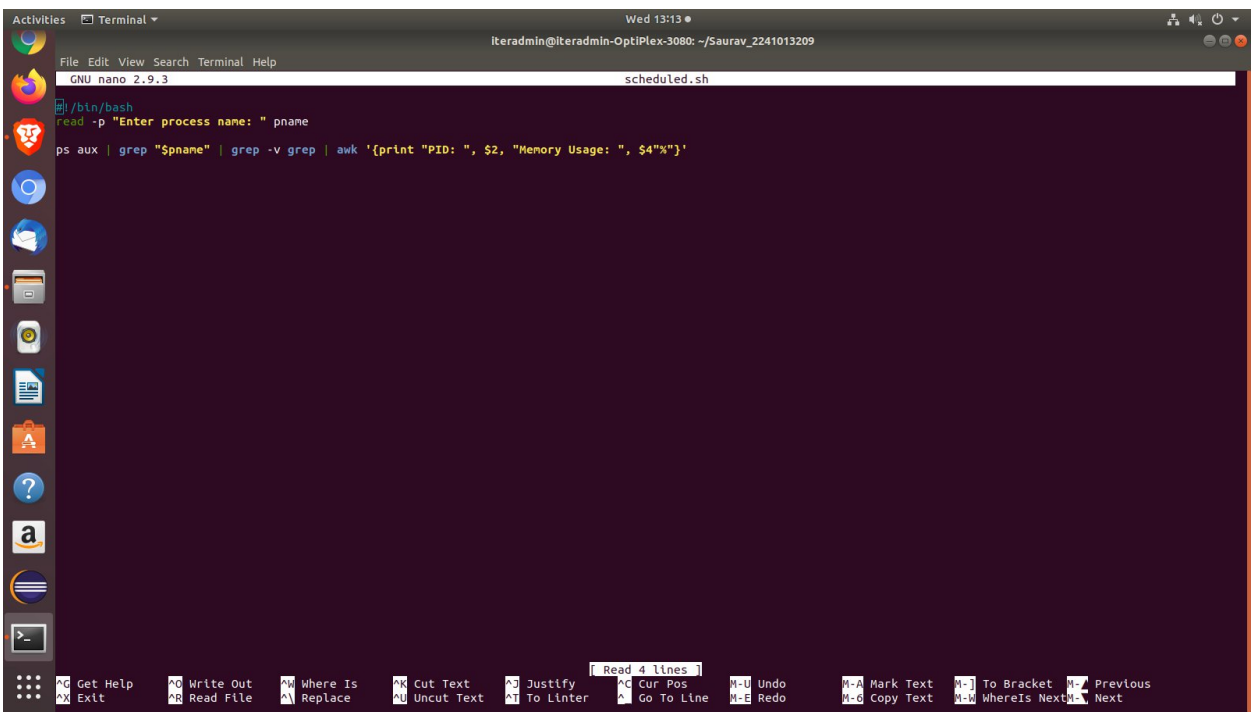


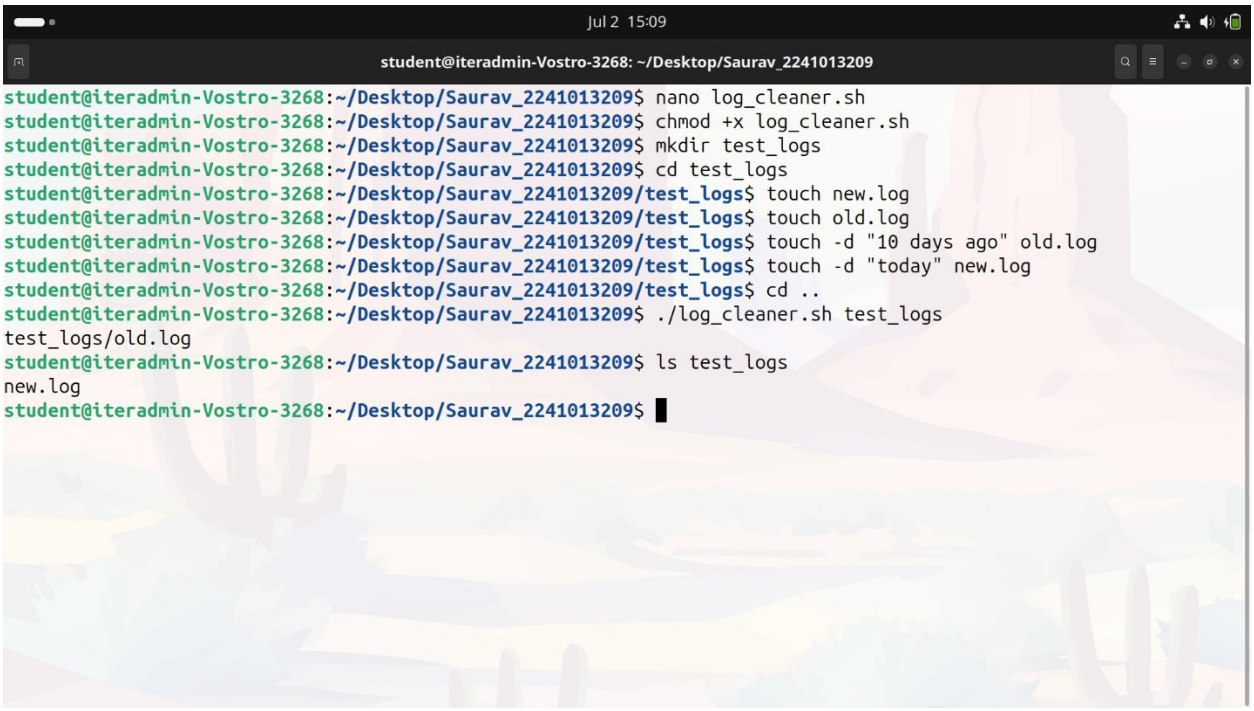


9. Process Monitor

Create a script that accepts a process name as input, checks if it is running using ps and grep, and displays its PID and memory usage.





10. Scheduled Cleanup

Write a script that deletes .log files older than 7 days from a given directory. Use find, conditionals, and optionally log what files were removed